Syllabus

Course Title: Capstone Writing: Multimodal Box Office Prediction and Film Investment Decision Support

Course Description

This self-directed capstone course focuses on expanding my current machine learning project, which predicts film box-office revenue. The goal of the project is to determine whether multimodal machine learning methods, integrating textual, categorical, and numerical features, can enhance the accuracy of box office predictions and aid in making informed investment decisions for films. Over the next six weeks, I will systematically expand the dataset, refine the modeling strategy, and prepare a final presentation for the capstone project.

Course Objectives

By the end of this course, I will:

1. Expand the existing course project into a more rigorous capstone study, using a stronger dataset and an improved methodology.
2. Address key limitations in the current project, including incomplete integration of structured metadata, limited handling of blockbuster outliers, and the lack of time-aware evaluation.
3. Compare baseline supervised models with multimodal approaches that combine text, metadata, and industry signals.
4. Frame the project's contribution in terms of how machine learning can serve as a practical decision-support tool in media investment contexts.
5. Deliver a polished final capstone presentation communicating the motivation, methods, results, and implications of the study.

Major Topics

- Refining the research question and capstone scope
- Data collection, cleaning, and dataset expansion
- Feature engineering across text, categorical, and numerical modalities
- Model comparison: baseline supervised models vs. multimodal extensions

- Error analysis, outlier handling, and time-aware evaluation
- Final capstone presentation preparation

Assignments and Deliverables

- Revised capstone research question and project plan
- Expanded and documented dataset
- Updated modeling pipeline with integrated features
- Model comparison results and error analysis
- Final capstone presentation

Tentative Weekly Schedule

Week 1: Capstone Framing and Data Planning

Review the existing course project and define the scope of the capstone extension. Clarify the research question and identify which additional data sources and variables will be incorporated (e.g., fuller metadata, cast/director features, review-related variables).

Week 2: Data Collection and Preprocessing

Collect, merge, clean, and document the expanded dataset. Resolve missing values, alignment issues, and variable formatting. Integrate numerical metadata and structured variables into the modeling pipeline, moving beyond the current TF-IDF-only approach.

Week 3: Baseline Model Revision

Rebuild and evaluate improved baseline models using the expanded feature space. Establish clear performance benchmarks for comparison with more advanced approaches.

Week 4: Advanced Multimodal Modeling

Test multimodal approaches that combine text, metadata, and other signals. Compare results against the revised baselines and assess which feature combinations contribute most to prediction accuracy.

Week 5: Error Analysis and Time-Aware Evaluation

Analyze model failures, with particular attention to blockbuster outliers and highly skewed outcomes. Implement a temporal evaluation design that better reflects real-world forecasting and investment screening conditions.

Week 6: Final Presentation Preparation and Submission

Synthesize all results and prepare the final capstone presentation, covering motivation, methods, key findings, limitations, and implications for film investment decision-making.